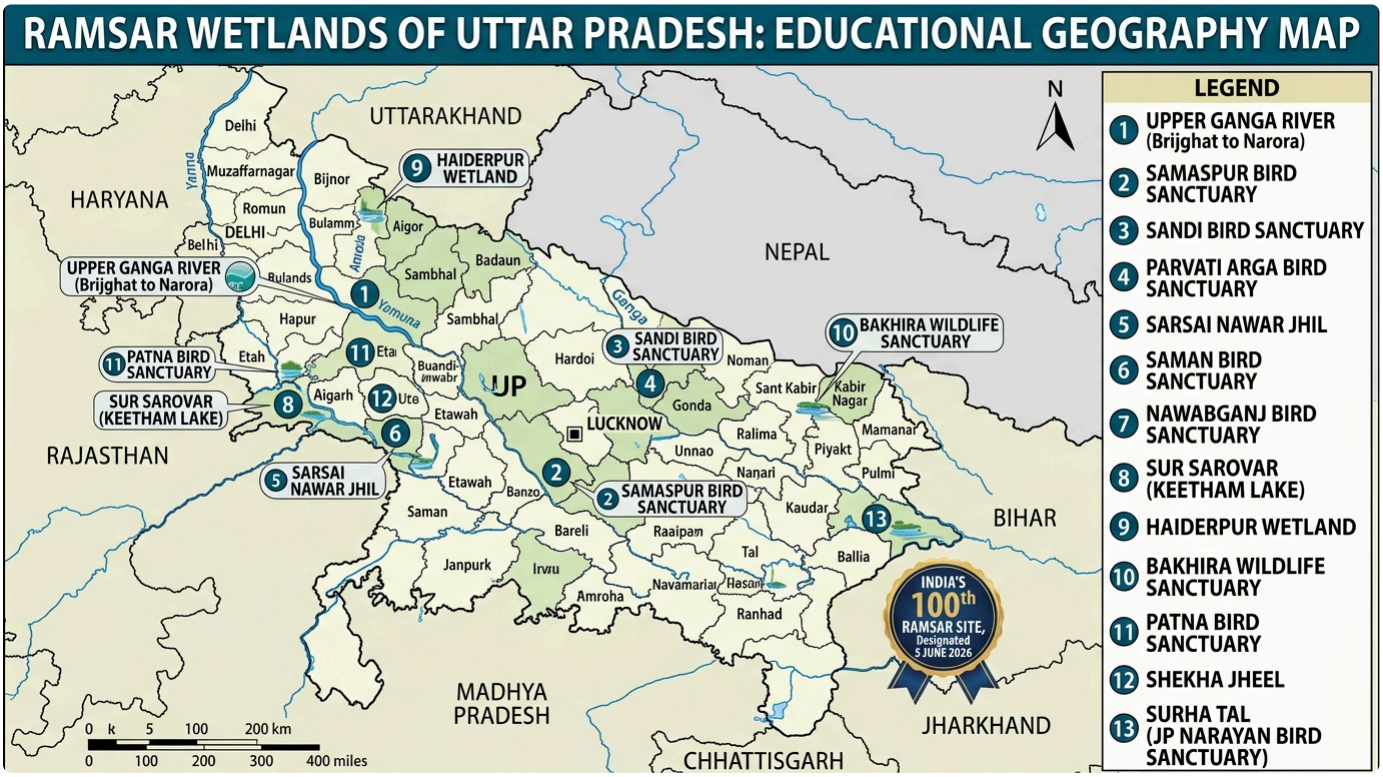
Topic 4 — Lakes, Waterfalls & Water Resources

Covers syllabus (click to expand)

Lakes: Lakes of India | Origin types (tectonic, glacial, oxbow, lagoon, crater, aeolian, landslide, artificial) | Lagoon | Saline (लवणीय) | Freshwater | Endorheic | Ramsar (रामसर) / UP wetlands **Waterfalls:** Formation (youth, knickpoint, hard–soft rock) | Major Waterfalls | Heights | River–state pairs **Dams, Projects & Canals:** Major Dams | Multipurpose | Ramganga (रामगंगा) | Tulbul | Tehri (टिहरी) | Bhakra Nangal (भाखड़ा नांगल) | Hirakud (हीराकुंड) | Sardar Sarovar (सरदार सरोवर) | Nagarjuna (नागार्जुन) Sagar (नागार्जुन सागर) | Major Canals | Hydroelectric | India–Bhutan Hydropower **Water resources:** Surface vs groundwater | Irrigation map | National Water Policy | PMKSY / Atal Jal / JJM / Namami Gange (नमामि गंगे) | Interlinking | Quality hazards

Sources baked in: NCERT Geography Class 11 (drainage / landforms), Class 12 (Water Resources), lake classification + waterfall + groundwater notes, Drishti / PW water-resources heads, **Ghatnachakra** Irrigation & Canals (CA–171–177), **Ghatnachakra** Multipurpose River Valley (CA–178–195), UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ — Chilika vs Sambhar vs Wular, Kunchikal ≠ Jog, dam–river / multipurpose (SSP–NBA, DVC 1948, Sidrapong), Ramganga specs, Bhutan hydro, UP Ramsar **Last verified:** August 2026 **Current Affairs:** India reached **100 Ramsar** sites on **5 Jun 2026**. UP has **13**. The 100th is **Surha Tal / JP Narayan BS, Ballia**.

Current Affairs (this topic)



Year	Fact	Why asked	Source
5 Jun 2026	India designates **100th Ramsar — Jai Prakash Narayan Bird Sanctuary (Surha Tal), Ballia, UP** (World Environment (पर्यावरण) Day (विश्व पर्यावरण दिवस))	"How many Ramsar in India / latest UP site?"	MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) / PIB
2025–26	UP Ramsar total **13** — adds **Patna (पटना) BS (Etah)**, **Sheikha/Shekha Jheel (Aligarh (अलीगढ़))**, **Surha Tal (Ballia)**	State-paper list + 2025 Q58 "NOT in UP"	MoEFCC LS annex
2025	India among top Ramsar countries in Asia; proposals in pipeline include Sauj Jheel (UP)	Count vs names trap	MoEFCC
Ongoing	**Jal Jeevan Mission** extended (rural tap water) — water-*supply* overlay, not a dam (दाम -)–river fact	Don't confuse with multipurpose dams	Jal Shakti (शक्ति)
2019–20	Six UP Ramsar added in the Jan 2020 window	Number question	UPPCS 2020

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q93

What is the correct sequence of lakes in descending order of size?

A. Victoria, Superior, Chilka, Michigan B. Superior, Victoria, Michigan, Chilka C. Michigan, Superior, Victoria, Chilka D. Chilka, Michigan, Superior, Victoria

Show answer

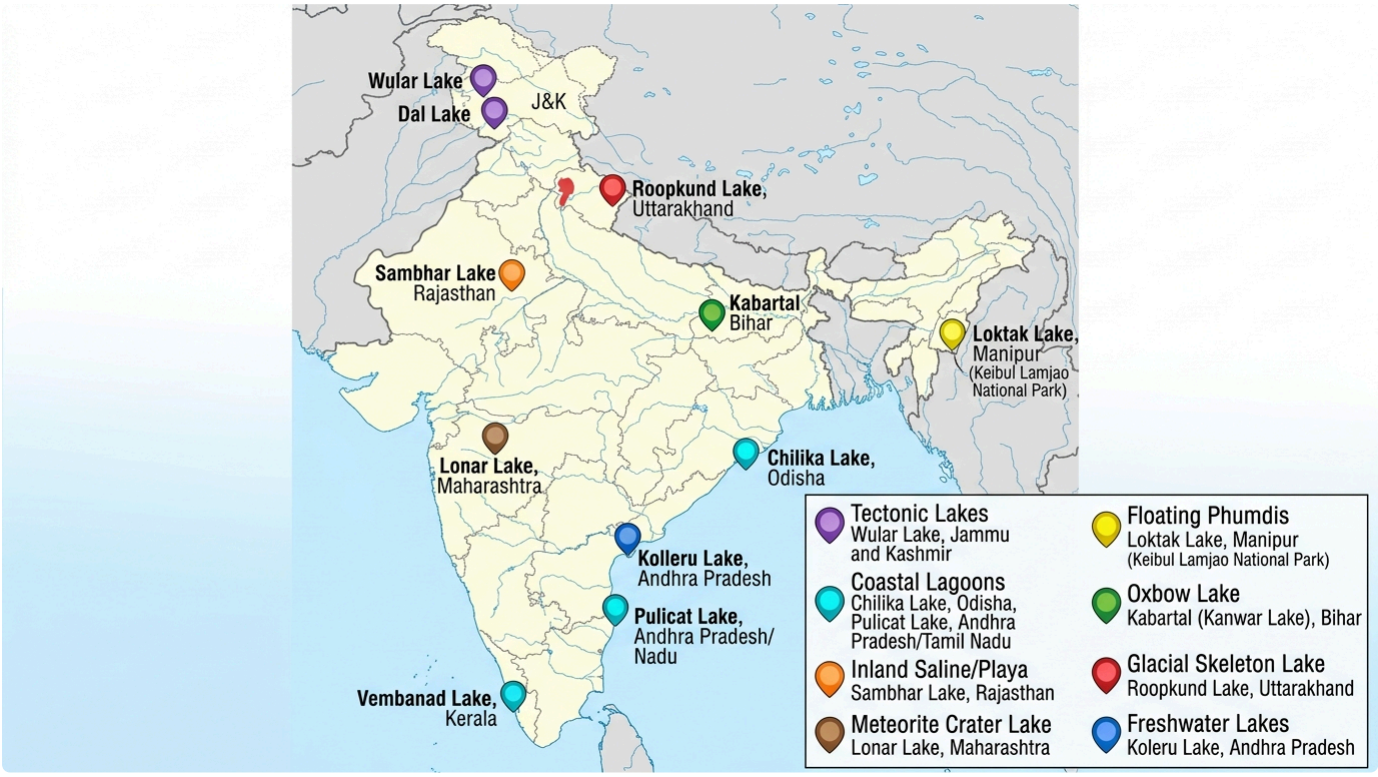
Logic: Superior vs Victoria decides the head of the sequence. **Ans: B (Series B provisional key).** Superior > Victoria > Michigan >> Chilka. Victoria-first is the tropical-lake trap.

N.0 Framework

- A **lake** is standing water in a basin on land. It is not the ocean, and it is larger and deeper than a pond.
- A **lagoon** is barred from the sea by a spit or bar. Notes therefore treat lagoons as **marine-deposit** lakes, not true inland lakes.
- An **endorheic** lake has no surface outlet. It loses water by evaporation or seepage and often turns saline.
- **Temporary** desert pans exist where evaporation exceeds precipitation. **Permanent** lakes are deep enough that precipitation and inflow beat evaporation.
- Most lakes on Earth are freshwater. Salt (लवणाध्यक्ष) lakes form where there is no outlet or where evaporation is fierce (Dead Sea, Great Salt Lake, Aral, Sambhar belt).
- Notes classify lakes by **how the hollow formed**, then by chemistry. UPPCS still asks the chemistry kings first (Chilika / Sambhar / Wular). Learn origin so a new question can switch the angle.

- No lake is permanent on a geological clock. Draining and silting will fill it.
- India's **water-resource** story is separate from "name the lake." Precipitation is large. Utilisable water is much smaller. Groundwater now irrigates more land than canals. Multipurpose dams store, generate, and cushion floods. They do not create new rain.

N.1 Lakes of India



Papers ask lakes in **two different ways**. Do not mix them in one mental list.

1. Chemistry / size kings (first question type)

Ask: *What kind of water is it, and who is the largest?*

If the question says...	Pick	Where	Why this one wins
Largest **lagoon** / east-coast lagoon	**Chilika**	Odisha	Barred from the sea; brackish; biggest east-coast lagoon
Largest **saline** (लवणीय) (**no** word "inland")	**Chilika**	Odisha	Papers often treat Chilika as the saline king by area
Largest **inland saline**	**Sambhar**	Rajasthan	Desert playa; no sea link; evaporation makes it salt
Largest **freshwater**	**Wular**	J&K (Jhelum)	Usual volume fact; tectonic basin

Do not swap

Wrong thought	Correct
"Saline" → always Sambhar	Only if the question says **inland**
Chilika = inland Rajasthan lake	Chilika = Odisha **coast**
Wular = saline	Wular = **fresh**
Kolleru = lagoon king	Kolleru = **freshwater** delta (डेल्टा) lake; Chilika is lagoon king
Pulicat = largest lagoon	Pulicat = **second** after Chilika

> **Logic:** "largest saline" (no inland) → **Chilika**. "Inland saline" → **Sambhar**. "East-coast lagoon" → **Chilika**.

2. Origin kings (second question type)

Ask: *How was the hollow made?*

Origin	Meaning in one line	Indian fact	Trap
Tectonic	Basin from earth movement / warp	**Wular** (also freshwater king)	Calling Wular a lagoon
Oxbow	Cut-off river meander on a plain	**Kabartal / Kanwar** (Bihar)	Calling it a coastal lagoon
Crater (meteorite)	Hollow from a meteor hit	**Lonar** (Buldhana, MH)	Calling it a volcano or RJ salt pan
Glacial / high mountain	Ice-cut tarn or high closed basin	Chandra Tal, Roopkund, Tsomgo; Pangong / Tso Moriri (high, often brackish)	Calling Pangong India's freshwater size-king
Lagoon (bar)	Sea water trapped behind a sandbar	**Chilika**, Pulicat	Calling Chilika inland saline
Wind / playa	Desert hollow, water evaporates	**Sambhar** belt	Using Sambhar for every "saline" question
Artificial	Reservoir behind a dam	Gobind Sagar, Hirakud, Tehri, Pant Sagar	Treating them as natural glacial lakes

Quick read: The size/chemistry table answers "largest saline / lagoon / freshwater." The origin table answers "oxbow / crater / tectonic / glacial." The same lake can appear in both — **Wular** is freshwater **and** tectonic; **Chilika** is a lagoon **and** bar-origin water body.

Origin types

Earth movement

- A **tectonic** lake sits in a crustal warp, fracture, or subsidence. **Wular** on the Jhelum is India's freshwater tectonic king. Huge world tectonic basins include **Titicaca** and the **Caspian**.

- **Pangong Tso** and **Tso Moriri** sit in high Ladakh basins tied to India–Eurasia collision. They are endorheic and brackish. Do not treat them as size-kings of Indian freshwater.
- A **rift-valley** lake is long, narrow and deep between parallel faults. East African examples (Tanganyika, Malawi, Turkana) and the **Dead Sea** (below sea level) are the world set.
- India has no Tanganyika-scale rift lake. Narmada and Tapi (तापी) occupy rifts, but they are **rivers**, not lakes.

Glaciation

- A **cirque / tarn** fills a hollow cut into a mountain ridge. **Chandra Tal** and **Suraj Tal** (Himachal), **Gurudongmar** and **Tsomgo / Changu** (Sikkim), and **Roopkund** (Garhwal) are the Himalayan tarn set.
- **Rock-hollow** glacial lakes are ice-scoured basins. Finland is the world "land of lakes." Canada and Siberia hold most Northern Hemisphere freshwater lakes.
- A **moraine-dammed** lake sits behind glacial debris. Unstable ice or debris dams can burst as a **GLOF**.

Volcanic and impact

- A **crater or caldera** lake fills a blown volcanic hollow. Krakatau is a world example.
- **Lonar** in Buldhana is a **meteorite** crater in Deccan (दक्कन) basalt, saline–alkaline, Ramsar 2020. Notes list it under crater lakes. Do **not** treat it as a volcanic caldera or a Rajasthan salt pan.

Erosion

- A **karst** lake fills a limestone solution hollow, or appears when a cavern roof collapses. They are local, not Indian size-kings.
- A **wind-deflated** hollow in a desert may tap groundwater and become a **playa**. The **Sambhar–Didwana–Pachpadra–Lunkaransar** belt is this family.

Deposition and barriers

- An **oxbow** forms when a meander neck is cut off. The Ganga plain north of the present channel is oxbow country.
- **Kabartal (Kanwar Lake)** in Begusarai is Asia's large freshwater oxbow and Bihar's first Ramsar site. It is rain-fed on the Burhi Gandak (बूढ़ी गंडक) belt.
- A **lagoon** is a marine-deposit lake behind a bar. **Chilika** is the Indian fact. **Pulicat** is second.
- A **landslide or dun (दून)** lake is a short-lived barrier lake. Outer Himalayan **duns** (including the Dehradun (देहरादून) hollow) were lakes a few centuries ago in the usual narrative.
- A **man-made** lake is a reservoir behind a dam. **Gobind Sagar**, **Hirakud**, **Tehri Lake**, **Pant Sagar**, and **Nagarjuna Sagar** reservoir are the set. The world analogue is Lake Mead above Hoover Dam.

Why lakes matter (lakes and man)

- Large lakes are inland waterways. The Great Lakes–St Lawrence (लॉरेंस) is the world teaching example.
- Lakes store water for towns and irrigation. **Kolleru** is a freshwater storage wetland on the Godavari–Krishna delta, not a lagoon king.
- Artificial lakes generate hydropower. **Hirakud** and **Bhakra** are the Indian teaching pair.

- Lakes moderate (नरम दल) local climate, grow fish, and attract tourists. Salt lakes yield common salt and, in some world basins, borax.
- They also silt, eutrophy, and shrink.

Lagoons and coastal wetlands

- A **lagoon** is a shallow coastal water body separated from the sea by a sandbar, spit, or barrier island that runs parallel to the shore.
- Kerala's lagoons and backwaters are also called **kayals** (Vembanad Kayal, Ashtamudi Kayal, Punnamada Kayal).
- The **Chilika** lagoon lies on the Odisha coast (Northern Circars belt), near the Mahanadi mouth. It is India's largest east-coast lagoon and often answers "largest saline" when the question omits **inland**.
- **Nalabana** is the bird island inside Chilika. An outer channel links the lagoon to the Bay of Bengal.
- Chilika is **not** on the west coast and **not** a Rajasthan salt lake.
- **Pulicat** straddles Andhra Pradesh and Tamil Nadu. It is the second-largest brackish / lagoon water body after Chilika.
- **Sriharikota** bars Pulicat from the Bay of Bengal. The Satish Dhawan space centre sits on that spit.
- **Vembanad** (also Vembanad Kayal / Vembanad–Kol) lies on Kerala's **west coast** (Malabar / southwest coast), separated from the Laccadive Sea by a narrow barrier. It borders **Alappuzha, Kottayam, and Ernakulam**.
- It is the **largest lake in Kerala** and the **longest lake in India** (~96 km).
- Do **not** call it India's overall largest lake — that crown splits by type.
- **Chilika** is the lagoon / often largest saline answer.
- **Wular** is the freshwater volume answer.
- The wider **Vembanad–Kol wetland** is often called India's largest wetland system on the southwest coast and a major **brackish** tropical wetland. It is a **Ramsar** (रामसर) site (2002).
- It is a **kayal** (Kerala backwater / lagoon-type water body), not an inland desert lake. The Nehru Trophy boat race and Kumarakom tourism belt are the culture facts.
- **Ashtamudi** ("eight branches") is a palm-shaped Kerala backwater Ramsar site in Kollam.
- **Sasthamkotta** is Kerala's largest **freshwater** lake (not Tamil Nadu — UPSC (संघ लोक सेवा आयोग) 2022 trap).
- **Kolleru** sits inland between the Godavari and Krishna deltas. Emergence left it freshwater. Do not crown it the lagoon king. Some papers treat it as India's largest freshwater lake **by area**; **Wular** remains the usual **volume / overall freshwater** fact.
- **Periyar Lake** (Kerala) is **artificial** (reservoir). It is **not** a lagoon (IAS 2002).
- **Kaliveli** in Tamil Nadu is the 2020 Q56 coastal match.
- Kerala, Odisha and West Bengal hold large lagoon and backwater tracts. NCERT notes the water is often brackish, yet it is used for fishing and some paddy and coconut.

Saline, crater and high-altitude lakes

- **Sambhar** lies across Jaipur, Ajmer (अजमेर) and Nagaur. It is India's largest **inland** saline lake.
- Sambhar salt pans and flamingos are the picture. It is Ramsar-listed.

- **Didwana**, **Kuchaman**, **Sargol**, **Khatu**, **Pachpadra** and **Lunkaransar** are Rajasthan saline / playa lakes (UPSC 2021 style name-set).
- **Pushkar** (Ajmer) is a small sacred **artificial** freshwater tank. It is not the saline king.
- **Nakki** is a tourism lake at **Mount Abu**, Rajasthan. It is not Sambhar.
- **Lonar** is the Maharashtra meteorite crater (Buldhana).
- **Pangong Tso** is a high-altitude brackish lake on the India–China line in Ladakh.
- **Tso Moriri** and **Tso Kar** are Ladakh's high Ramsar / brackish set.
- **Yaya Tso** in Ladakh is called a "bird's paradise." In 2023 it was declared Ladakh's first Biodiversity Heritage Site under the Biological Diversity Act. It is **not** fed by the Sutlej.

Freshwater icons and regional belts

- **Wular** in Jammu and Kashmir is India's largest freshwater lake **by the usual volume fact**. The Jhelum flows through it.
- Tulbul is the navigation and storage project on the Wular–Jhelum belt.
- **Dal** in Srinagar (Zabarwan belt) is the houseboat and tourism lake. It is **not** the size king.
- **Anchar** is another Jammu and Kashmir lake in the match set.
- **Loktak** in Manipur is famous for **phumdis** (floating biomass).
- **Keibul Lamjao** is the only floating national park. It holds the **Sangai** (सांगाइ) brow-antlered deer.
- The Kumaon (कुमाऊँ) belt holds **Nainital (नैनीताल), Bhimtal, Sattal and Naukuchiatal**.
- **Roopkund** in Uttarakhand is a glacial tarn called the **Mystery Lake / Skeleton Lake** because of human skeletons found on its shore. It is **not** in Himachal Pradesh.
- **Ghepan** is a snow-covered glacial lake in the **Lahaul** valley of Himachal Pradesh.
- **Renuka** in Himachal is a Ramsar wetland lake. **Nako** is another Himachal high lake.
- **Pong / Maharana Pratap Sagar** is the Beas reservoir-wetland in Himachal.
- **Harike** in Punjab sits at the Sutlej–Beas meeting.
- **Hokera** Wetland is in **Jammu and Kashmir**, not Punjab (UPSC 2022).
- **Deepor Beel** is Assam's Ramsar wetland near Guwahati. **Chapanala**, **Chandubi**, and **Haflong** are other Assam lake names in PCS lists.
- **Phulhar / Gomat Taal** near Madho Tanda (**Pilibhit, UP**) is the source of the **Gomti**.
- **Bhojtal** (Upper Lake) at Bhopal is a large urban (नगरीय) reservoir-wetland.
- **Hussain Sagar** and **Usman Sagar** at Hyderabad (हैदराबाद) are artificial.
- **Chembarambakkam** (south of Chennai, Tamil Nadu) feeds the **Adyar** and supplies Chennai water.
- **Kodaikanal** lake (Tamil Nadu) is **artificial** (UPSC 2018). Renuka and Nainital are natural; Kolleru is natural freshwater.
- **Surajkund** (सूरजकुंड) is in **Haryana** (match-list fact with Ashtamudi–Kerala, Pulicat–TN/AP, Roopkund–UK).
- **Sukhna** at Chandigarh is a man-made siltation-sensitive lake.
- Rajasthan's **Pichola** and **Fateh Sagar** are tourism lakes. They are not Sambhar.
- **Kabartal (Kanwar Lake)** in Begusarai is Asia's large freshwater oxbow and Bihar's only Ramsar site. **Telhar Kund** waterfall is in **Kaimur** district, Bihar (Rohtas (रोहतास) Plateau belt).

> **Logic:** Keibul Lamjao National Park (केइबुल लामजाओ) sits on **Loktak**, not Wular, Dal, or Kolleru.

Ramsar Convention (short fact)

- The **Ramsar Convention** (adopted (अंगीकृत) **1971** at Ramsar, Iran; in force **1975**) is the global treaty for wetland conservation. It is the only global environmental treaty focused on one ecosystem (पारिस्थितिकी तंत्र) type.
- India's Ramsar count and UP list live in the UP wetlands card below; do not confuse state sites with the treaty year.

Lake–state match bank

Next year can pick **any** high-yield pair. Learn the bank; the 2020 set is marked.

Row order in a match table is not the answer code.

Lake	State / UT	Note
Sala ★2020	**Arunachal Pradesh**	2020 Q56
Badkhal ★2020	**Haryana** (Faridabad belt)	2020 Q56
Loktak ★2020	**Manipur**	Phumdis; Keibul Lamjao
Kaliveli ★2020	**Tamil Nadu**	Coastal wetland
Chilika	**Odisha** (ओडिशा)	Largest E-coast lagoon
Pulicat	**Andhra Pradesh + Tamil Nadu**	Sriharikota bar
Vembanad	**Kerala**	Largest in Kerala; longest in India; **west coast**
Ashtamudi	**Kerala**	Kayal / Ramsar
Sasthamkotta	**Kerala**	Largest freshwater in Kerala (not TN)
Periyar Lake	**Kerala**	Artificial; **not** a lagoon
Kolleru	**Andhra Pradesh**	Freshwater; between Godavari–Krishna
Wular	**J&K**	Largest freshwater (volume)
Dal	**J&K** (Srinagar)	Tourism; not size king
Anchar	**J&K**	Match distractor set
Hokera	**J&K**	**Not** Punjab
Sambhar	**Rajasthan** (थार)	Largest inland saline
Didwana / Kuchaman / Pushkar / Nakki	**Rajasthan**	Nakki = Mount Abu

Lake	State / UT	Note
Pichola / Fateh Sagar	**Rajasthan** (Udaipur (उदयपुर))	Tourism; not Sambhar
Lonar	**Maharashtra** (Buldhana)	Meteorite crater
Kabartal / Kanwar	**Bihar** (Begusarai)	Oxbow Ramsar
Deepor Beel / Chapanala	**Assam**	Deepor = Guwahati Ramsar
Renuka / Nako / Ghepan / Pong	**Himachal Pradesh**	Renuka = Ramsar; Ghepan = Lahaul
Roopkund / Nainital / Bhimtal	**Uttarakhand** (उत्तराखंड)	Roopkund = Mystery Lake (**not** HP)
Surajkund / Sultanpur	**Haryana**	Sultanpur **not** UP
Rudrasagar	**Tripura**	**Not** UP
Pangong / Tso Moriri / Yaya Tso	**Ladakh** (लद्दाख)	High / brackish; Yaya Tso = BHS
Tsomgo / Gurudongmar	**Sikkim**	Glacial tarns
Hussain Sagar / Usman Sagar	**Telangana** (Hyderabad)	Artificial
Chembarambakkam / Kodaikanal	**Tamil Nadu**	Kodaikanal = artificial
Bhojtal (Upper Lake)	**Madhya Pradesh** (Bhopal)	Urban wetland
Nalsarovar	**Gujarat**	Bird sanctuary / Ramsar class
Harike	**Punjab**	Sutlej–Beas meet
Sukhna	**Chandigarh**	Man-made
Phulhar / Gomat Taal	**Uttar Pradesh** (Pilibhit)	Gomti source

World lake–country match bank

Later papers can swap in Ladoga, Victoria, Titicaca, and similar names — learn the bank.

Lake	Country	tag / trap
Baikal ★2023	**Russia** (Siberia)	Deepest + oldest freshwater; huge volume
Michigan ★2023	**USA** only	Only Great Lake wholly in USA
Onega ★2023	**Russia**	Europe's 2nd-largest; **not Canada**
Maracaibo ★2023	**Venezuela**	Large brackish; oil belt
Ladoga	**Russia**	Europe's **largest** lake (Onega is 2nd)
Caspian Sea	Russia / Kazakhstan / Turkmenistan / Iran / Azerbaijan	World's **largest lake by area**; saline
Dead Sea	Israel / Jordan	Rift; hypersaline; below sea level

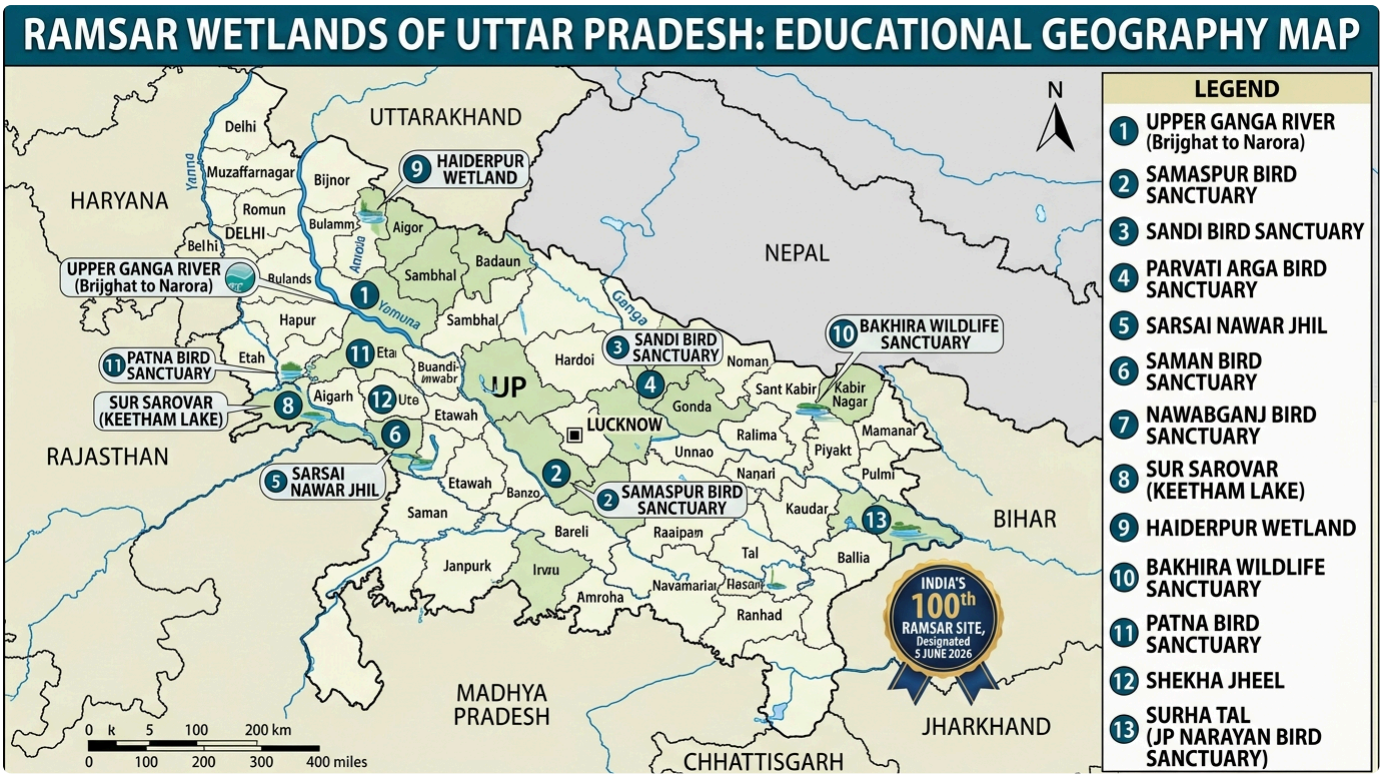
Lake	Country	tag / trap
Superior	**USA + Canada**	Largest Great Lake by area
Huron / Erie / Ontario	**USA + Canada**	Shared Great Lakes (unlike Michigan)
Victoria	Uganda / Kenya / Tanzania	Africa's largest by area
Tanganyika	Several E. African states	Deep rift lake
Malawi (Nyasa)	Malawi / Mozambique / Tanzania	Rift lake
Turkana	Kenya (mainly)	Rift; saline
Titicaca	**Peru + Bolivia**	Highest large navigable lake
Great Bear / Great Slave	**Canada**	Not Onega
Loch Ness / Loch Lomond	**Scotland (UK)**	
Eyre	**Australia**	Endorheic salt lake
Chad	Chad / neighbours	Shrinking inland
Aral	Kazakhstan / Uzbekistan	Shrinking saline

Superlatives (once)

Crown	Lake
Largest lake by **area** (world)	**Caspian** (saline)
Largest **freshwater** by **area** (usual set)	**Superior**
Largest / deepest freshwater by **volume**	**Baikal**
Lowest / densest salt	**Dead Sea**

> **Logic:** ** Onega and Ladoga are both **Russia**. Do not park Onega in Canada. Michigan is USA-only; Superior is shared.

UP wetlands (state paper gold)



Ramsar **in UP** (learn the names that *are* UP). UP total **13** as of June 2026.

Site	District / belt
Upper Ganga River	Brijghat to Narora
Nawabganj	Unnao
Parvati Arga	Gonda
Saman	Mainpuri
Samaspur	Rae Bareli
Sandi	Hardoi
Sarsai Nawar	Etawah
Sur Sarovar	Agra
Haiderpur	Bijnor–Muzaffarnagar (Ganga)
Bakhira	Sant Kabir (कबीर) Nagar
Patna Bird Sanctuary	Etah
Sheikha / Shekha Jheel	Aligarh
Jai Prakash Narayan BS / Surha Tal	**Ballia** (India's 100th Ramsar, 5 Jun 2026)
Name that sounds like UP	Actual state
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Site	District / belt
Rudrasagar	**Tripura**
Sultanpur	**Haryana**
Surinsar–Mansar	**J&K**

> **Logic:** ** “Which Ramsar sites are **NOT** in UP?” — pick Rudrasagar, Sultanpur, Surinsar–Mansar; do not throw out Sarsai Nawar / Samaspur.

Artificial lakes (reservoirs)

- **Gobind Sagar** is the Bhakra reservoir on the Sutlej.
- **Hirakud** reservoir is on the Mahanadi.
- **Tehri Lake** is on the Bhagirathi.
- **Pant Sagar** is on the Rihand (रिहंद) in Uttar Pradesh.
- **Nagarjuna Sagar** reservoir is on the Krishna.
- **Indira Sagar** lake on the Narmada in Madhya Pradesh is often cited as India’s largest reservoir **by volume**.
- These are lakes of **dam origin**.

PYQ — UPPCS Prelims 2021, Q74

Largest saline water lake in India is

A. Chilka

B. Sambhar

C. Lonar

D. Wular

Show answer

Ans: A — no “inland” in the question → Chilika by area. Sambhar is the inland-saline trap.

N.2 Waterfalls



- A **waterfall** is a steep or vertical drop of a stream. Falls may be natural or artificial. Most Indian natural falls sit in mountain / plateau country with hard–soft rock contrasts.
- Soft rock erodes faster than hard rock. The hard ledge remains as a lip and water plunges at the **knickpoint**.
- Most falls sit in the **youth** stage of a river. A fall in base level or rejuvenation also creates knickpoints.
- A **plunge** fall drops in one sheet. A **cascade / tiered** fall steps down in many drops. A **segmented** fall splits into side-by-side streams.
- **Famous is not the same as tallest.** Height lists shift with databases. Fact **relative (सापेक्ष) order**, **river–state pairs**, and the UPPCS height key.

Superlative	Fact	Trap
Highest	**Kunchikal** (Varahi, Karnataka; older figure ~455 m)	Jog
Tallest **plunge** (newer World Waterfalls Database)	**Nohkalikai** (Meghalaya (मेघालय), ~340 m)	Using Nohkalikai to beat every Kunchikal height question
Largest by **width**	**Jog / Gersoppa** (Sharavati, Karnataka)	Calling Jog highest
Largest by **volume**	**Shivanasamudra** (Cauvery, Karnataka)	Putting volume crown on Jog
Waterfall	~Height	River
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Superlative	Fact	Trap
Kunchikal	older ~455 m; newer DB ~183 m	**Varahi**
Barehipani	~399 / ~217 m by DB revision	Budhabalanga
Nohkalikai	~340 m	Plateau stream
Nohsngithiang (Seven Sisters) / Mawsmai	~315 m class	—
Dudhsagar	~310–320 m	**Mandovi**
Barkana	~259 m	Sita (सीता) belt
Jog (Gersoppa)	~253 m	**Sharavati**
Khandadhar	~244 m	Korapani Nala → Brahmani
Palani	—	Kullu belt
Hundru	~98 m	**Subarnarekha**
Jonha	—	**Rarhu**
Dasam / Dasong	—	**Kanchi** (कांची)
Lodh / Budha Ghagh	~143 m (468 ft)	**Burha**
Dhuandhar	~30 m	**Narmada**
Kapildhara	—	**Narmada**
Dugdhdhara / Bheraghat belt	—	**Narmada**
Bhalkund (Rahatgarh)	—	**Bina**
Chachai	~130 m	Bihad / Beehar (Tamsa (तमसा))
Chulia	—	**Chambal** (चंबल)
Joranda	~150 m	Budhabalanga
Duduma	—	**Machhkund**
Gokak	—	**Ghataprabha**
Shivanasamudra	~90–98 m	**Cauvery**
Hogenakkal	—	**Cauvery**
Athirappilly	—	Chalakudy
Chitrakote	wide, low (~29 m)	**Indravati**
Amritdhara	—	**Hasdeo (Hasdo)**
Kevti / Rakim Kund	—	Rewa plateau streams

- In the UPPCS **2018** height list, **Kunchikal** is India's highest waterfall. Newer databases may rank other falls higher for specific measures, but they do not make **Jog** the highest.
- **Kunchikal** is a cascade on the **Varahi**. Mani Dam has reduced its dry-season flow.
- **Jog** is segmented on the **Sharavati**. Sharavati is a **west-flowing** Karnataka river to the Arabian Sea at Honnavar. Jog is **not** India's highest; it is the renown / width fact.
- Some older notes once called Sharavati a **Cauvery tributary**. That is **wrong**. Jog is not on the Cauvery.
- **Linganamakki** dam stores Sharavati water for Karnataka hydropower.
- **Nohkalikai** is the tallest **single-drop plunge** in newer lists. For overall height, **Kunchikal** remains the usual textbook answer; for plunge height alone, use **Nohkalikai**.
- **Dudhsagar** is in **Goa** in the UPPCS **2022** answer, even though the Mandovi belt touches Karnataka.
- **Dhuandhar** cuts marble at Bhedaghat on the Narmada.
- **Hundru** drops off the Ranchi plateau on the Subarnarekha.
- **Lodh / Budha Ghagh** on the **Burha** is Jharkhand's highest fall. Pairing it with Kanchi or Barakar is wrong.
- **Shivanasamudra** on the Cauvery is the **volume** king and hosted India's early hydro station (**1902**).
- **Kapildhara** is on the **Narmada** near Amarkantak. Pairing it with the Godavari is a classic unmatched trap.
- **Chitrakote** on the **Indravati** (Bastar) is the "Niagara of India."
- Rewa district tourism still places **Chachai** on the **Bihad / Beehar**, a Tamsa tributary. The **2021 official answer** treated **Chachai–Bihad** as the unmatched pair.

Match fodder: Dudhsagar–Mandovi · Duduma–Machhkund · Gokak–Ghataprabha · Jog–Sharavati.

PYQ — UPPCS Prelims 2018, Q24

Which of the following is the highest waterfall in India?

A. Jog fall

B. Kunchikal fall

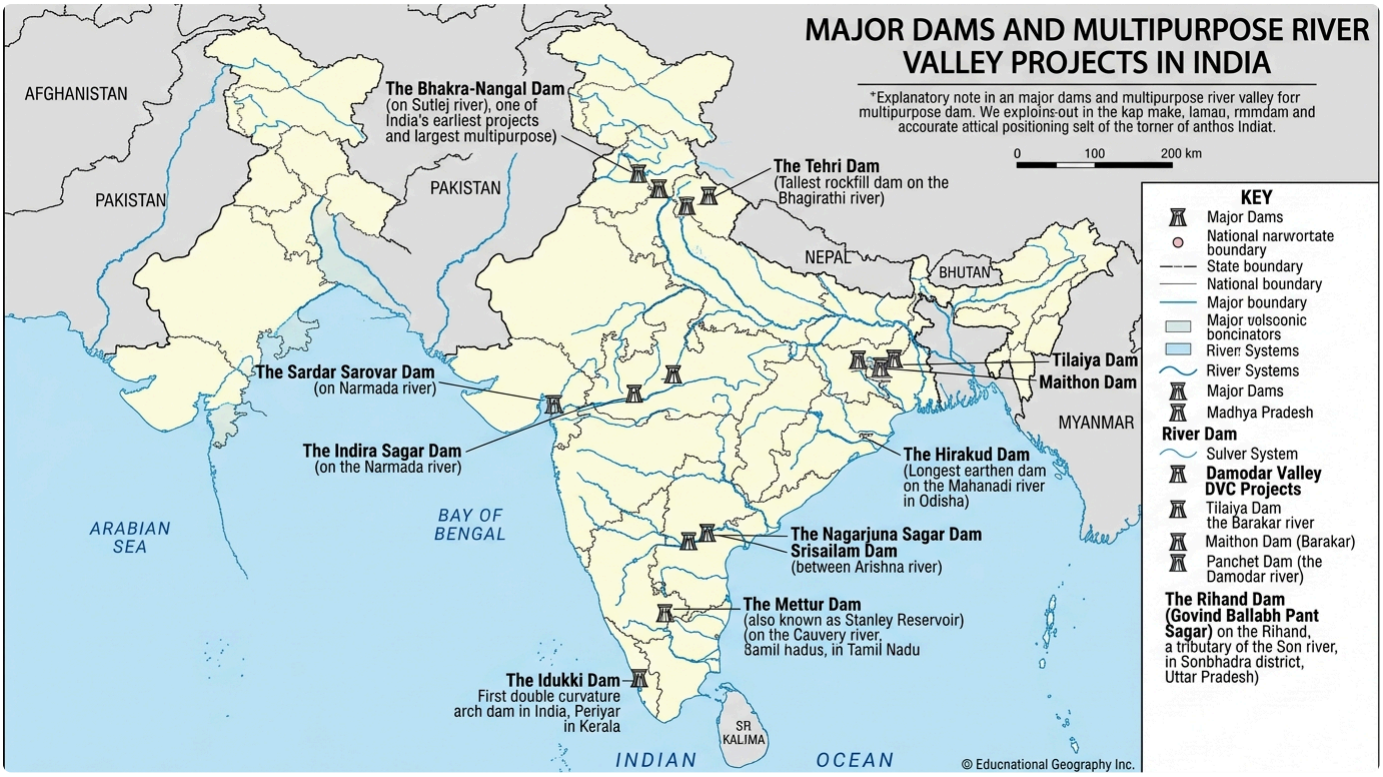
C. Rakim Kund fall

D. Kevti fall

Show answer

Logic: Famous ≠ tallest. UPPCS **2018** followed the older height list (Kunchikal). **Ans: B** — Kunchikal (Varahi, KA). Jog is the renown / width trap.

N.3 Dams, canals and hydropower



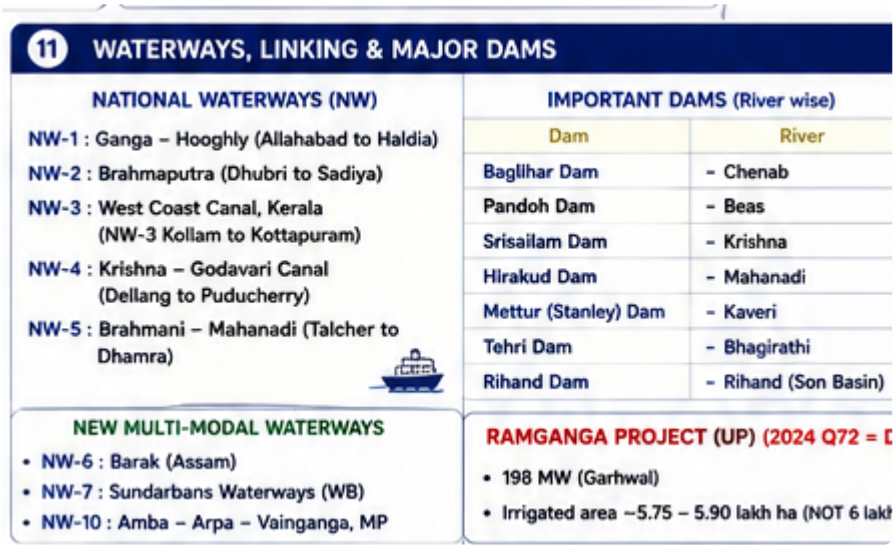
- A **dam** stores water in a reservoir. A **barrage** diverts with little storage. **Farakka** and **Nangal** headworks are barrages, not storage twins of Bhakra.
- **Multipurpose** means irrigation plus power plus flood cushion, and sometimes drinking water / navigation. Nehru called multipurpose river-valley projects the **“temples of modern India.”**
- NCERT’s teaching set of river-valley projects is **Bhakra–Nangal, Hirakud, Damodar Valley, Nagarjuna Sagar, and the Indira Gandhi Canal**.
- Dam type matters. **Bhakra** is a concrete **gravity** dam. **Tehri** is **rock-fill / earth-fill** in a seismic Himalayan valley. **Idukki** is a **double-curvature arch** dam between the Kuravan and Kurathi hills.
- India’s **oldest** hydro station is **Sidrapong** near **Darjeeling** (**1897**). The **second** oldest is **Shivasamudram** on the **Cauvery** in Karnataka (**1902**) — power-focused, not the irrigation king.

Star projects

Project	River	Where / type	Reservoir / extra
Bhakra–Nangal	**Sutlej**	HP dam + Punjab headworks; **gravity**; joint **PB–HR–RJ** (command also touches HP)	**Gobind Sagar**; completed **1963**; Nehru’s “temple of modern India” tag
Hirakud	**Mahanadi**	Odisha (~15 km from Sambalpur); started **1948**, completed **1957**	Longest earthen-dam narrative (~**4.8 km** main / ~**26 km** with dykes); ~10 lakh ha irrigation note
Sardar Sarovar	**Narmada**	Near **Navagam**, Gujarat; height ~**163 m**; ~**1450 MW**	Beneficiaries **GJ, RJ, MH, MP**; MP takes the largest Narmada water share; **NBA / Medha Patkar** opposed height rise

Project	River	Where / type	Reservoir / extra
Indira Sagar (Narmada Sagar)	**Narmada**	Near **Punasa / Khandwa**, MP	Largest reservoir **by volume**; ~1.23 lakh ha + ~**1000 MW** target; **Harsud** town submerged (**2004**)
Nagarjuna Sagar	**Krishna**	AP–TS border; completed **1967**	Multipurpose; Nalgonda–Khammam–Guntur–Prakasam belt; height ~**125 m**
Tehri	**Bhagirathi** (+ **Bhilangana** confluence at old Tehri)	Uttarakhand; **rock-fill**; THDC	India's **tallest** dam (~**260.5 m**); cleared **1972**, work from **1978**; seismic / displacement protest

Master Table — Major Dams & Multipurpose Projects in India (भारत के प्रमुख बांध एवं बहुउद्देशीय परियोजनाएं)



11 WATERWAYS, LINKING & MAJOR DAMS

NATIONAL WATERWAYS (NW)

NW-1 : Ganga – Hooghly (Allahabad to Haldia)
NW-2 : Brahmaputra (Dhubri to Sadiya)
NW-3 : West Coast Canal, Kerala (NW-3 Kollam to Kottapuram)
NW-4 : Krishna – Godavari Canal (Dellang to Puducherry)
NW-5 : Brahmani – Mahanadi (Talcher to Dhamra)

NEW MULTI-MODAL WATERWAYS

- NW-6 : Barak (Assam)
- NW-7 : Sundarbans Waterways (WB)
- NW-10 : Amba – Arpa – Vainganga, MP

IMPORTANT DAMS (River wise)

Dam	River
Baglihar Dam	- Chenab
Pandoh Dam	- Beas
Srisailam Dam	- Krishna
Hirakud Dam	- Mahanadi
Mettur (Stanley) Dam	- Kaveri
Tehri Dam	- Bhagirathi
Rihand Dam	- Rihand (Son Basin)

RAMGANGA PROJECT (UP) (2024 Q72 = D

- 198 MW (Garhwal)
- Irrigated area ~5.75 – 5.90 lakh ha (NOT 6 lakh

(High-yield for match-the-column with Rivers, States, and Specific Features)

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Tehri Dam (टिहरी बांध)	**Bhagirathi** (+ Bhilangana confluence)	Uttarakhand	**Tallest dam in India** (260.5 m); rock-fill earth dam; THDC
Bhakra-Nangal (भाखड़ा नांगल)	**Sutlej**	Himachal Pradesh & Punjab	**Highest concrete straight gravity dam** (226 m); reservoir: **Gobind Sagar**; joint venture of PB, HR, RJ
Hirakud Dam (हीराकुंड बांध)	**Mahanadi**	Odisha (Sambalpur)	**Longest earthen dam in the world** (main dam 4.8 km; total length with dykes 25.8 km); Gandhi Minar & Nehru Minar

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Sardar Sarovar Dam	**Narmada**	Gujarat (Navagam, Kevadia)	Height 163 m; benefits Gujarat, Rajasthan, Maharashtra, MP; Statue of Unity downstream; **Narmada Bachao Andolan** (Medha Patkar)
Indira Sagar Dam (Narmada Sagar)	**Narmada**	Madhya Pradesh (Khandwa)	**Largest water reservoir in India by storage volume** (12.22 BCM); submerged Harsud town
Nagarjuna Sagar Dam	**Krishna**	Andhra Pradesh & Telangana	World's largest masonry dam; built 1955–1967; named after Buddhist scholar Acharya Nagarjuna
Srisailam Dam	**Krishna**	Andhra Pradesh & Telangana	Second largest capacity working hydroelectric station; located in deep gorge of Nallamala hills
Almatti Dam (Lal Bahadur Shastri)	**Krishna**	Karnataka (Bijapur)	Main reservoir of Upper Krishna Irrigation Project
Pandoh Dam (पंडोह बांध)	**Beas**	Himachal Pradesh (Mandi)	Diverts Beas waters into Sutlej via Beas-Sutlej Link for power at Dehar
Pong Dam (Maharana Pratap Sagar)	**Beas**	Himachal Pradesh (Kangra)	Earth-fill embankment dam; wetland Ramsar site
Baglihar Dam (बगलीहार)	**Chenab**	Jammu & Kashmir (Ramban)	Run-of-the-river project; cleared after neutral expert Raymond Lafitte adjudication under Indus Waters Treaty
Salal Dam (सलाल बांध)	**Chenab**	Jammu & Kashmir (Reasi)	Run-of-the-river hydroelectric project on Chenab rock pedestal
Dulhasti Dam (दुलहस्ती)	**Chenab**	Jammu & Kashmir (Kishtwar)	Run-of-the-river power project on Chandra/Chenab river
Pakal Dul & Ratle	**Chenab** & Marusudar	Jammu & Kashmir (Kishtwar)	Major new hydropower projects under execution
Uri Dam (Uri-I & II)	**Jhelum**	Jammu & Kashmir (Baramulla)	Near LoC; run-of-the-river hydro plant
Tulbul Navigation Project	**Jhelum** (Wular Lake outlet)	Jammu & Kashmir	Navigation barrage contested by Pakistan; suspended since 1987
Rihand Dam (Govind Ballabh Pant)	**Rihand** (tributary of Son)	Uttar Pradesh (Pipri, Sonbhadra)	**Largest man-made artificial lake by area in India**; power source for Renukoot aluminium plant

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Matatila Dam (माताटीला बांध)	**Betwa**	Uttar Pradesh (Lalitpur)	Multipurpose dam; joint project with MP
Rajghat Dam (Rani Laxmibai)	**Betwa**	UP & MP border (Lalitpur/Chanderi)	Inter-state multipurpose project on Betwa
Parichha Dam (परीछा बांध)	**Betwa**	Uttar Pradesh (Jhansi)	Thermal power station water supply & canal irrigation
Chambal Project (4-stage cascade)	**Chambal**	MP & Rajasthan	1. **Gandhi Sagar** (Mandsaur, MP) 2. **Rana Pratap Sagar** (Rawatbhata, RJ) 3. **Jawahar Sagar** (Kota, RJ) 4. **Kota Barrage** (irrigation canals)
Idukki Dam (इडुक्की बांध)	**Periyar**	Kerala	Concrete **double-curvature arch dam** between Kuravan & Kurathi hills; underground power house at Moolamattom
Mullaperiyar Dam	**Periyar**	Kerala (leased to Tamil Nadu)	Masonry gravity dam built 1895 by British engineer John Pennycuick; administered by TN
Mettur Dam (Stanley Reservoir)	**Cauvery**	Tamil Nadu (Salem)	Built 1934; chief irrigation reservoir for Cauvery delta farmers
Krishna Raja Sagara (KRS)	**Cauvery**	Karnataka (Mandya/Mysuru)	Designed by Sir **M. Visvesvaraya**; Brindavan Gardens located below dam
Ukai Dam (Vallabh Sagar)	**Tapi**	Gujarat (Surat)	Second largest reservoir in Gujarat after Sardar Sarovar
Kakrapar Project	**Tapi**	Gujarat	Weir project for irrigation + Kakrapar Atomic Power Station
Koyna Dam (कोयना बांध)	**Koyna** (tributary of Krishna)	Maharashtra (Satara)	Largest completed hydroelectric power plant in Maharashtra; triggered **1967 Koynanagar earthquake**
Damodar Valley Project (DVC)	**Damodar & Barakar**	Jharkhand & West Bengal	First multipurpose river valley project of independent India (1948); modeled on US **Tennessee Valley Authority (TVA)**; dams at **Tilaiya, Maithon, Belpahari** (on Barakar) and **Panchet** (on Damodar); converted Damodar from "Valley of Sorrow" to "Valley of Plenty"
Tungabhadra Project	**Tungabhadra** (Krishna trib.)	Karnataka & Andhra Pradesh	Major multipurpose joint project (dam at Mallapuram, 2441 m long, 49.3 m high); powerhouses at Mallapuram & Hampi

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Pochampad Project (Sri Rama Sagar)	**Godavari**	Telangana (Nizamabad)	Major lifeline irrigation project for North Telangana
Jayakwadi Project	**Godavari**	Maharashtra (Paithan, Chhatrapati Sambhajinagar)	Huge earthen dam across Godavari; reservoir named **Nath Sagar**
Lower Sileru & Upper Sileru	**Sileru** (Godavari basin)	Andhra Pradesh & Odisha	Major hydel stations on the AP-Odisha border
Sabarigiri Project	**Pamba & Kakki**	Kerala (Pathanamthitta)	Second largest hydroelectric project in Kerala
Sharavati Project	**Sharavati**	Karnataka (Jog Falls / Gersoppa)	Major hydroelectric project at Mahatma Gandhi Hydroelectric Works
Ghataprabha & Malaprabha	**Ghataprabha & Malaprabha**	Karnataka (Belagavi / Bagalkote)	Irrigation reservoirs in the Krishna basin
Bhima Project	**Bhima / Pawana**	Maharashtra (Ujjani Dam, Solapur)	Multipurpose irrigation and hydel project
Banasagar Project	**Son**	Madhya Pradesh, UP & Bihar	Joint interstate project (shared cost and benefit 2:1:1 by MP, UP, Bihar) at Deolond (Shahdol)
Thein Dam (Ranjit Sagar Dam)	**Ravi**	Punjab & Jammu & Kashmir	Highest earth-fill gravity dam in Punjab; hydel and irrigation
Mayurakshi Project (Canada Dam)	**Mayurakshi**	West Bengal & Jharkhand	Dam at Massanjore (built with Canadian assistance)
Kangsabati Project	**Kangsabati & Kumari**	West Bengal (Bankura/Purulia)	Earthen dam for flood control and irrigation
Papanasam Project	**Thamirabarani**	Tamil Nadu (Tirunelveli)	Hydroelectric power station
Kundah Hydroelectric Project	**Kundah & Bhavani**	Tamil Nadu (Nilgiris)	Multi-stage cascade hydel system built with Canadian aid
Loktak Hydroelectric Project	**Manipur River / Loktak Lake**	Manipur (Churachandpur)	Power generation and lift irrigation via Loktak Lake trans-basin tunnel
Sharda Sahayak Project	**Ghaghara & Sharda**	Uttar Pradesh	Major canal diversion feeder system for eastern/central UP plains

Multipurpose valley cards (Ghatnachakra)

Narmada Valley Project

- Planned as one of the world's largest valley programmes: about **30 major**, **135 medium**, and **3,000** small dams / barrages on the Narmada and tributaries.
- Mega dams include **Sardar Sarovar (SSP)** in Gujarat and **Indira Sagar / Narmada Sagar** in MP.
- Cascade in MP also includes **Bargi, Omkareshwar (ओंकारेश्वर), Maheshwar**, plus smaller projects such as **Tawa, Barna, Kolar, Sukta, Matiyari** (Tawa ~**13.5 MW**; Hoshangabad belt between Satpura (सतपुड़ा) NP and Bori WLS).
- Tributary projects include **Man**, **Chandra Shekhar Azad** (चंद्रशेखर आज़ाद) (Hathani), and **Upper Veda (वेद)**.
- **Ban Sagar** is on the **Son** — **not** a Narmada dam. Do not put it in the Narmada list with Bargi / Omkareshwar / Indira Sagar.
- **Narmada Bachao Andolan** grew when Medha Patkar organised project-affected people from **1986**.
- Local groups merged as **NBA** in **1989**.
- The movement opposed raising SSP height.
- The Supreme Court stressed rehabilitation.

Bhakra–Nangal

- Joint venture of **Punjab, Haryana and Rajasthan** on the **Sutlej**.
- Command also expands into **Himachal** (हिमाचल).
- Two linked works stand at **Bhakra** and **Nangal**.
- The reservoir is **Gobind Sagar**.

Cauvery hydro & dispute

- **Shivasamudram** (**1902**) is the second-oldest Indian hydro work.
- The **KRS (Krishna Raja Sagara)** profile is often tagged to **M. Visvesvaraya**.
- **Cauvery River Water Dispute** parties are **Tamil Nadu, Karnataka, Kerala and Puducherry** (not Gujarat).

Chambal Valley Project

- This is a joint **Madhya Pradesh–Rajasthan** chain on the Chambal.
- The Chambal rises near **Janapav / Mhow** and joins the Yamuna near Etawah.
- **Gandhi Sagar** (MP, Mandsaur) came first (**1957–60**).
- **Rana Pratap Sagar** (RJ, Rawatbhata) follows.
- **Jawahar Sagar** (RJ) and then **Kota Barrage** complete the chain.
- **Gobind Sagar** (Bhakra) and **Govind Ballabh Pant Sagar** (Rihand, Sonbhadra (सोनभद्र)) are **not** Chambal dams.

Damodar Valley (DVC)

- Damodar is the **"Sorrow of Bengal"** — fast upper course, sluggish lower course, floods and silt.
- **DVC** was established in **1948** as the **first multipurpose river-valley project of independent India**.
- **Tilaiya**, **Maithon** and **Belpahari** stand on the **Barakar**.

- **Konar** stands on the Konar.
- **Panchet** stands on the Damodar.
- Canals from **Durgapur Barrage** irrigate the Howrah–Hooghly–Bardhaman belt.

Other high-yield projects

Project	River / place	Note
Telugu Ganga	Krishna → **Chennai** drinking water	Joint TN–AP–KA–MH
Chukha	Wang Chu / Raidak, **Bhutan**	India-aided; handed to Bhutan **1991**; **~336 MW**
Mayurakshi	Mayurakshi; Canada / Massanjore dam in JH	Shared **WB–JH**
Babli	Godavari, **Maharashtra**	Disputed with Telangana / AP
Mahakali Treaty	Sharda (शारदा) / Mahakali	India–**Nepal** (**1996**); Sharda, Tanakpur, Pancheshwar
Gandak Project	Gandak Barrage (**1959** treaty)	India–Nepal; **UP + Bihar** (+ Nepal)
Kishau	**Tons** (टौंस) (Yamuna)	HP–UK border; benefits HP/UK (+ others)
Kalpong	Kalpong	First hydro of **Andaman (अंडमान) & Nicobar (निकोबार)**
Rihand / Pant Sagar	Rihand (Son tributary), Pipri, Sonbhadra	Largest UP multipurpose note; Obra power below dam
Rajghat / Rani Lakshmibai (रानी लक्ष्मीबाई)	**Betwa** (बेतवा), Lalitpur	Joint UP–MP
Matatila	Betwa	Joint UP–MP
Pochampad	Godavari	Telangana
Tapovan–Vishnugad	**Dhauliganga** (धौलीगंगा), Chamoli (चमोली) UK	
Dulhasti	**Chenab** (चिनाब), Kishtwar	
Baglihar	**Chenab**, Ramban	IWT / World Bank dispute note
Suil	Suil (Ravi tributary), Chamba HP	
Kol Dam	**Sutlej**, HP	NTPC; silt check for Bhakra note
Meja (RJ)	**Kothari** (Banas), Bhilwara	Irrigation — not UP's Meja
Kalpasar	Gulf of Khambhat, **Gujarat**	Freshwater reservoir / tidal idea
Teesta hydro	Teesta	Sikkim / West Bengal (TLDP III–IV = WB)

Project	River / place	Note
Malampuzha	Malampuzha (Bharathapuzha), Kerala	Largest Kerala reservoir note
Aliyar / Isapur / Kangsabati	—	All are **water reservoirs**

Interstate water-dispute tribunals (match fodder)

Tribunal	Year / note	Parties (count)
Mahadayi	**2010**	Goa, Karnataka, Maharashtra (**3**)
Godavari (गोदावरी) (Bachawat)	**April 1969**	AP, MP, Chhattisgarh, Odisha, Karnataka (**5**)
Narmada	**6 Oct 1969**	Gujarat, MP, Maharashtra, Rajasthan (**4**)
Mahanadi	**2018**	Odisha, Chhattisgarh (**2**)

- Large-dam count descending order (CGPCS 2021): **Madhya Pradesh > Gujarat > Rajasthan > Telangana > Andhra Pradesh**. Absolute (निरपेक्ष) "biggest dams" note also tags **Maharashtra, Madhya Pradesh, Gujarat**.

River-wise dam map (match fodder)

Indus (सिंधु) system

- **Pandoh** is on the **Beas** (ब्यास). It is not on the Ravi.
- **Pong / Maharana Pratap Sagar** is also on the Beas in Himachal (bird sanctuary reservoir).
- **Ranjit Sagar (Thein)** is on the **Ravi** (रावी).
- **Baglihar**, **Salal** and **Dulhasti** are on the **Chenab**.
- **Nathpa Jhakri** and **Kol Dam** are Sutlej hydro works in Himachal.

Ganga–Yamuna–UP

- **Tehri** and **Koteshwar** stand on the Bhagirathi (Bhilangana joins at the Tehri site).
- **Rihand** (रिहंद) dam makes **Govind Ballabh Pant Sagar** at Pipri, Sonbhadra.
- **Matatila** and **Rajghat (Rani Lakshmibai)** are Betwa projects serving Bundelkhand.
- **Narora** is a Ganga **barrage**, not a high dam.
- **Kalagarh** is the site of the **Ramganga Dam**.

Narmada–Tapi–Chambal

- Narmada cascade in MP includes **Bargi, Indira Sagar, Omkareshwar, Maheshwar**, then **Sardar Sarovar** in Gujarat.
- **Gandhi Sagar**, **Rana Pratap Sagar**, **Jawahar Sagar** and **Kota Barrage** are the Chambal chain (**MP–RJ**).
- **Ukai** and **Kakrapar** are on the **Tapi** (तापी) in Gujarat.
- **Ban Sagar** is on the **Son**, not the Narmada.

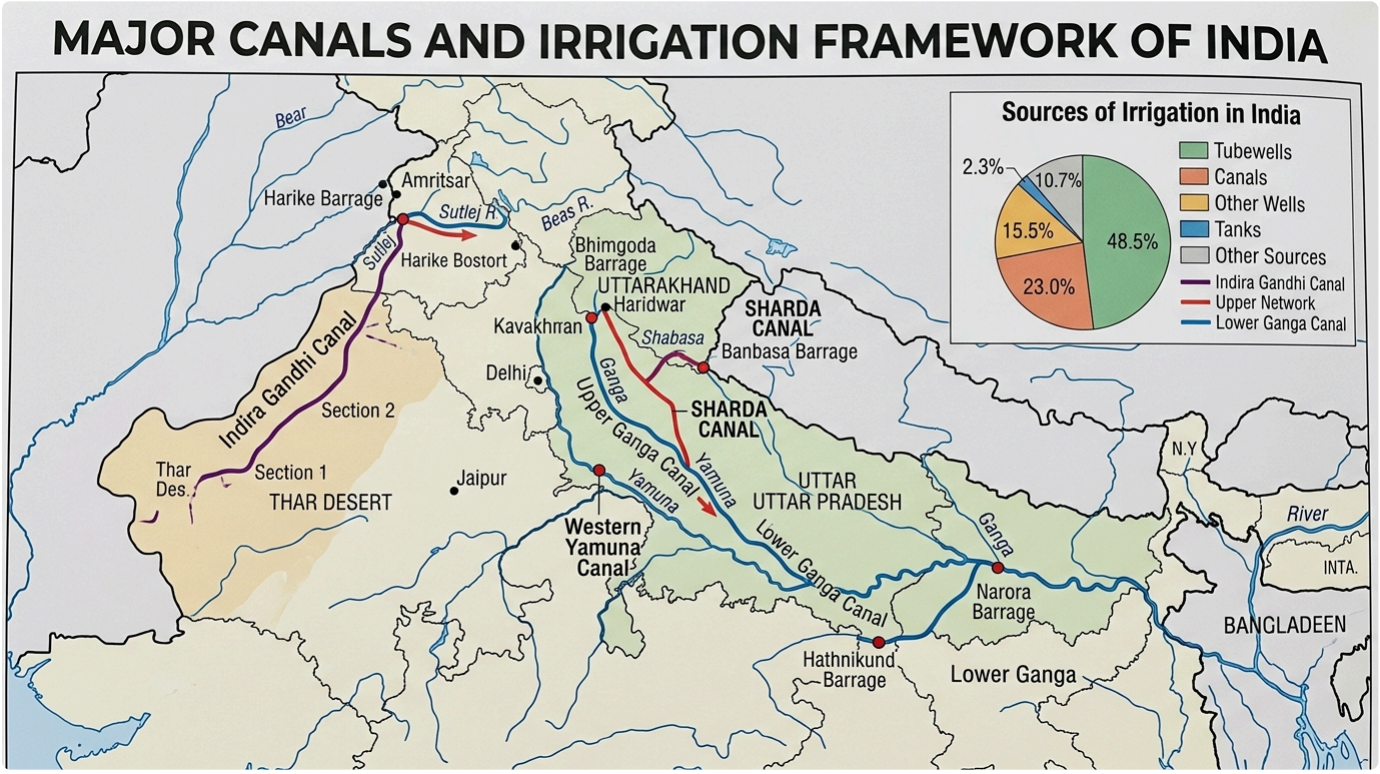
East-flowing peninsula

- **Srisailam** and **Almatti** and **Nagarjuna Sagar** are on the **Krishna**.
- **Tungabhadra (तुंगभद्रा) Dam** is on the Tungabhadra (KA–AP joint).
- **Jayakwadi** and **Pochampad** are on the Godavari.
- **Nizam (निजाम) Sagar** is on the **Manjra**, a Godavari tributary.
- **Hirakud** is on the Mahanadi. **Rengali** is on the Brahmani.
- **Mettur** (Stanley Reservoir) is on the Cauvery in Tamil Nadu.
- **Krishna Raja Sagara (KRS)** and **Shivasamudram** are on the Cauvery in Karnataka.

West-flowing Ghats and others

- **Koyna** is on the Koyna (Krishna system) in Maharashtra. The **1967** shock is the textbook **reservoir-induced seismicity** case.
- **Idukki** / **Cheruthoni** are on the **Periyar** in Kerala.
- **Mullaperiyar** is also on the Periyar. Kerala hosts the dam. Tamil Nadu operates it.
- **DVC** dams **Tilaiya, Konar, Maithon, Panchet** sit on the Damodar–Barakar system.

Canals



Canal	Source / offtake	Region	Tag
Gang Canal	Sutlej at **Hussainiwala** (Firozpur)	North Rajasthan / Bikaner command	Among world's **oldest developed** systems; **1927**; Maharaja **Ganga Singh**
Indira Gandhi Canal	**Harike Barrage** (Sutlej–Beas confluence, Punjab)	Ganganagar, Bikaner, Jodhpur, Jaisalmer	India's **longest** (~**649 km** main); foundation **Mar 1958** (G.B. Pant); also draws Beas–Ravi–Sutlej waters (Pong / Beas project support)

Canal	Source / offtake	Region	Tag
Upper Ganga Canal	Ganga at **Bhimgoda, Haridwar** (right bank)	W-UP Doab (Haridwar–Saharanpur–Muzaffarnagar–Meerut (मेरठ)–Ghaziabad–Bulandshahr–Aligarh–Etah–Mathura (मथुरा)–Firozabad–Mainpuri–Agra belt)	Father often tagged **Sir P.T. Cautley**; first discharge **8 Apr 1854**; main ~**298 km**; system ~**6496 km**; also feeds Delhi (~200 cusec) and Harduaganj TPP (~100 cusec)
Lower Ganga Canal	Ganga at **Narora** (Bulandshahr)	Central / lower Doab	Completed **1878**; main ~**99 km**; system ~**8278 km**
Eastern Yamuna Canal	Yamuna left bank at **Tajewala**; **Hathinikund** barrage ~3 km upstream (Haryana)	Hindon–Yamuna Doab: Saharanpur, Muzaffarnagar, Meerut, Ghaziabad	Revived functioning from **1830**; main ~**197 km**
Agra Canal	Yamuna (Okhla)	Agra–Mathura belt	
Sharda Canal	Sharda right bank near **Banbasa** (Nainital / UK border)	Protective irrigation across Rohilkhand–Awadh (अवध)–Purvanchal (पूर्वांचल) (Pilibhit to Prayagraj (प्रयागराज) belt)	Main ~**44 km**; system ~**9961 km** (longest UP canal-system length fact)
Sharda Sahayak / CAD	Sharda command	E / central UP	CADP projects also include Ramganga and Gandak
Betwa Canal	Betwa	Bundelkhand	
Bhakra / Sirhind	Sutlej system	Punjab–Haryana	
Buckingham Canal	East-coast backwaters	AP (Kakinada) → TN (Viluppuram); Coromandel	Freshwater **navigation** canal (~796 km), not IGC
Gandak / **Saran** canals	Gandak barrage near **Triveni / Balmiki Nagar** (~740 m)	NW Bihar	Saran irrigation canal drawn from this system
Ukai Canal	Tapti (ताप्ती) / Ukai	Gujarat	
Ghataprabha Canal	Ghataprabha	Karnataka	
Nira Canal	Nira (Bhima tributary)	Maharashtra	
Jawai Project	Jawai (Luni (लूनी) tributary)	**Rajasthan**	Trap: not Tamil Nadu
Mangalam	Bharathapuzha system	**Palakkad, Kerala**	One of several projects on that river

- Canal irrigation dominates the Indo-Gangetic plain.

- The peninsula was historically stronger on **tanks and ponds** because hard rock makes canals and wells hard to dig and most peninsular rivers are **seasonal**.
- Groundwater now covers more than **60%** of India's irrigated area.
- In **2018–19**, tubewells alone covered about **48.5%**.
- Wells plus tubewells covered about **64%**.
- Canals covered about **23%**.
- Tanks covered about **2.3%**.
- Do **not** say canals are the dominant national source.
- **Ghaggar** (घग्गर) water is **not** a source of the Indira Gandhi Canal.

Irrigation project classes (CCA)

Class	Culturable Command Area	Share of irrigation potential
Minor	Up to **2,000 ha** (wells, tubewells, tanks, lift, drip, sprinkler)	About **62%**
Medium	**2,000–10,000 ha**	Part of the remaining ~**38%** with major
Major	Above **10,000 ha**	Major + medium together about **38%**

Hydroelectric geography

- Himalaya (हिमालय) combines high **head** and perennial **flow**. Potential is enormous. Installed use is still a fraction.
- The peninsula uses monsoon storage plus Western Ghat head (Koyna, Idukki, Sharavati, Srisailam).
- Much of the annual flow of Godavari, Krishna and Kaveri (कावेरी) is already harnessed. NCERT says the Ganga and Brahmaputra still have unused surplus.
- **2025 Q31 fact:** Himalayan hydro potential is huge. Eastern UP and Bihar cannot fully tap the ranges immediately to their north because those mountains lie largely in **Nepal**. Both statements are true.

India–Bhutan hydropower

- Export-to-India is the pillar.
- Ratta names are **Chukha, Tala, Kurichhu and Mangdechhu (Mungdechhu)**.
- **2019 Q115:** the plant in the PM Modi Bhutan-visit context is **Mungdechhu / Mangdechhu**.
- **Punatsangchhu** is the delayed news name. It is not the 2019 fact.
- These are **Bhutanese** plants. They are not Indian dams.

PYQ — UPPCS Prelims 2025, Q31

With reference to hydroelectric power in Northern India, which of the following statements is/are correct?

1. The Himalayan region has enormous hydroelectric potential.
2. Eastern Uttar Pradesh and Bihar cannot utilize this potential because the Himalayas situated to their north lie within the territory of Nepal.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: C — Nepal geography is the limiter, not “no Himalayan rivers.”

N.4 Water resources (NCERT Class 12 + schemes)

- Water is a **cyclic abiotic** resource. Freshwater on Earth is only about **3%** of all water. Usable freshwater is a thinner slice still.

How much India has

- India has about **2.45%** of the world’s land, about **4%** of the world’s water, and more than **17%** of the world’s people (NCERT; many notes round people to **18%** and land to **2.4%**).
- Annual precipitation is about **4,000 cubic km**. Surface flow plus replenishable groundwater is about **1,869 cubic km**.
- Only about **60%** of that 1,869 is utilisable. Total utilisable water is about **1,122–1,123 cubic km**.
- Utilisable **surface** water is about **690 cubic km**. Replenishable **groundwater** is about **432–433 cubic km**.
- Surface sources are rivers, lakes, ponds and tanks. India has more than **10,000** rivers and tributaries longer than 1.6 km.

Uneven geography

- The **Ganga, Brahmaputra and Barak** basins cover about **one-third** of India’s area but hold about **60%** of surface water resources.
- North-west India and the Deccan are rainfall-deficient. Even wet Bengal and Bihar need irrigation when the monsoon breaks.

Who uses the water

- Irrigation dominates demand. NCERT: agriculture uses about **89%** of surface water and about **92%** of groundwater withdrawn.
- Industry uses about **2%** of surface water and about **5%** of groundwater. Domestic use is higher on the surface-water side (about **9%**).
- Future shares of industry and cities will rise. Irrigation will still be the giant.

Irrigation map

- Most of India sits in a **sub-tropical / high-temperature** belt (except western Himalaya, parts of the north-east, and some southern tracts). High temperature drives high evaporation, so sown crops need irrigation to cover the moisture deficit. Rainfall also varies in space and time. Rice, sugarcane and jute need a lot of water. HYVs need a regular moisture supply.

- In **2018–19 (P)**, tubewells covered about **48.51%** of national irrigated area.
- Wells plus tubewells together covered about **63.94%**.
- Canals covered about **22.96%**.
- Tanks covered about **2.33%**.
- The rest used other means.
- The dominant source is **wells and tubewells**, not canals alone.
- Punjab, Haryana and western Uttar Pradesh irrigate more than **85%** of net sown area.
- That is why the Green Revolution (हरित क्रांति) stuck there.
- **Haryana and Punjab** lead **% cropped area under irrigation**.
- **Uttar Pradesh** leads **absolute** tubewell / well irrigated area.
- In Punjab about **76%** of net irrigated area is wells and tubewells.
- In Haryana the well share is about **51%**.
- Over-pumping has dropped the water table.
- Dynamic groundwater resources available for irrigation (bcm order): **Uttar Pradesh ~40.72** > **Punjab ~26.69** > **Madhya Pradesh ~17.39** > **Maharashtra ~15.29** (West Bengal and Assam much lower among map options).
- NCERT MCQ fact: among south Indian states, **Tamil Nadu** has the highest groundwater utilisation of its potential.
- Chhattisgarh, Odisha and Kerala still use only a small share of groundwater potential.
- Gujarat, Uttar Pradesh, Bihar, Tripura and Maharashtra sit in the moderate band.
- **Protective / life-saving irrigation** means irrigation at **permanent wilting point (PWP)** when soil moisture can no longer sustain the plant.
- **Micro-irrigation** (drip / sprinkler) cuts fertiliser / nutrient loss and can check groundwater decline in some areas.
- It is **not** the only means of irrigation in dryland farming.
- **Drip** advantages include less weed, less soil erosion and higher water-use efficiency — not a guarantee of lower soil salinity as an automatic result.

Quality hazards

- **Fluoride** rises where over-withdrawal is intense in Rajasthan and parts of Maharashtra.
- **Arsenic** contaminates groundwater in parts of West Bengal and Bihar (Ganga belt).
- Intensive canal irrigation in Punjab, Haryana and western UP spreads **usar / salinity**.
- Organic and bacterial load is the main river-pollution story. The **Yamuna** (यमुना) between Delhi and Etawah is NCERT's most-polluted stretch.
- Other NCERT dirty stretches include Sabarmati (साबरमती) at Ahmedabad (अहमदाबाद), Gomti at Lucknow (लखनऊ), Ganga at Kanpur (कानपुर) and Varanasi (वाराणसी), Musi at Hyderabad, and the Cooum–Adyar in Chennai.
- The Water (Prevention and Control of Pollution) Act **1974** and the Environment (Protection) Act **1986** are the legal frame. Implementation has been weak.

National Water Policy 2012

- NWP 2012 wants a **national water framework law** and better interstate-river legislation.
- Safe **drinking water and sanitation** come first. Food security, poor farmers, and minimum ecosystem needs follow. After that, water is an **economic good**.
- The policy asks for water **footprints**, water **auditing**, climate-proof design, and community-managed services.
- Falkenmark thumb rule used in Year Book notes: below **1,700 m³** per person per year is **water stress**. Below **1,000 m³** is **scarcity**.

Schemes and conservation (keep Ken–Betwa short; drainage chapter owns the link)

- **Ministry of Jal Shakti** (2019) merged water resources with drinking-water and sanitation.
- **PMKSY** was approved on **1 July 2015** to cut monsoon dependence and take irrigation toward every farm (**Har khet ko pani** + **more crop per drop**). AIBP funds major irrigation. HKKP funds minor irrigation, water-body repair, and command-area work.
- Till **1 April 2008**, Rural Development ran three watershed programmes — **Integrated Wastelands Development Programme (IWDP)**, **Drought (सूखा) Prone Areas Programme (DPAP)**, and **Desert Development Programme (DDP)**. From 2008 they sit under **Integrated Watershed Management (वाटरशेड (जलग्रहण) प्रबंधन) Programme (IWMP)** / later WDC–PMKSY, under the **Department of Land Resources (MoRD)** — not Agriculture or Environment as the pair trap.
- **Hariyali** was launched by Prime Minister **Atal Bihari Vajpayee** on **27 January 2003** for rural watershed / water storage and wasteland-linked conservation through panchayats.
- **Command Area Development Programme (CADP)** (from **Dec 1974**) improves utilisation inside an existing command (field channels, drains, warabandi, levelling, cropping pattern). Classic UP projects: **Sharda Tributary**, **Ramganga**, **Gandak** (गंडक).
- **Garland Canal System** was proposed by Captain **Dinshaw J. Dastur**. The idea of linking rivers is first tagged to irrigation engineer **M. Visvesvaraya**; later **K.L. Rao** added another linking proposal. Both styles were later set aside by the water ministry in favour of other planning frames.
- **Pattiseema** Lift Irrigation links **Godavari** to **Krishna** (Andhra Pradesh).
- **Atal Bhujal Yojana** works in water-stressed gram panchayats of **Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh**.
- **Jal Jeevan Mission** is rural tap water. It is not a dam–river fact.
- **Namami Gange** (नमामि गंगे) cleans the Ganga. It is not a lake-origin fact.
- **Ken–Betwa** is the only National Perspective Plan link under implementation (Bundelkhand, MP–UP). Full specs live in the Drainage chapter.
- **Watershed** work stores runoff and recharges wells. Benefits include soil-runoff control, rainwater harvesting / recharge, and regeneration of vegetation — **not** national perennial–seasonal river linking.
- NCERT also names Andhra's **Neeru-Meeru** and Alwar's **Arvari Pani Sansad**.
- **Ralegan Siddhi** (Ahmednagar) is NCERT's watershed success story.
- **Rainwater harvesting** captures roof and surface rain to use or to recharge. It dilutes fluoride and nitrate, checks erosion, and can slow coastal salinity intrusion.

- Rajasthan's **kund / tanka** is a covered underground tank. **Johads** are village ponds. Tamil Nadu made rooftop harvesting compulsory in buildings.
- Recycle grey water for gardens and industry cooling. Keep the cleanest water for drinking.
- Desalination and river-linking are costly extras. Pricing and conservation still matter more for households.
- Agriculture has the **largest water-consumption share** among sectors in India.
